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Comparative Topology of Multiple Onco-PPI Networks

Complex Networks

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1 Introduction

Cancer is a complex disease in which multiple molecular alterations affect the normal functioning of the cell. These alterations do not act in isolation, but usually involve related proteins through functional or physical interactions. Therefore, the study of protein-protein interaction networks, known as PPIs, enables us to analyze cancer from a systemic perspective, considering not only genes or proteins individually, but the global organization of their interactions.

In this project, three PPI networks associated with different cancer types are studied: breast cancer, lung cancer, and ovarian cancer. Each network is represented as a graph where nodes correspond to proteins and edges correspond to interactions between them. Through this representation, several complex networks tools are applied to characterize the structure of the system and evaluate possible topological differences between the three cancer types.

The analysis includes the macroscopic characterization of networks considering measures such as number of nodes, edges, average degree, clustering coefficient, assortativity, and other measures. Moreover, the degree distribution is studied to determine if the networks present a heterogeneous structure with the presence of hubs or highly connected proteins. These proteins can play a relevant role in the network organization and the propagation of alterations inside the system.

Microscopically, properties are also analyzed through several measures such as centrality, with the objective of identifying the most potentially important proteins for the connectivity, influence, or strategic positioning in the network. Subsequently, the robustness of the network is studied through a percolation analysis comparing the effect of random failures with targeted attacks to nodes with high degree or high betweenness. Finally, community detection algorithms have been applied to identify modules of highly connected proteins using methods such as Louvain, Leiden, greedy, Infomap, and a Bayesian approach based on the DCSBM model.

FALTA ACABAR: mirar que mes hem posat

epidemic spreading?

Overall, this project seeks to compare the organization of the different PPI networks associated with different cancer types and explore the structural characteristics that may be related to the robustness, modularity, and relative importance of several proteins inside each network

2 Data Source

The project uses data from the STRING database [1], given that it provides protein-protein interaction networks for *Homo sapiens*. First, the global human network was downloaded and filtered to keep only high-confidence interactions, using a cutoff threshold of 0.7. Thus, connections between proteins were only considered when STRING assigned a sufficiently high confidence score.

Subsequently, instead of analyzing the whole human network, the project focuses on three cancer types: breast cancer, lung cancer, and ovarian cancer. To do so, the STRING app [2] within Cytoscape [3] was used to import each network as a disease query, entering the corresponding cancer name. A maximum number of 2000 proteins was also set for each cancer type.

Finally, each network was exported in `.sif` format, which has the following simple structure:

```
protein1    pp    protein2
```

where `protein1` and `protein2` are proteins and `pp` indicates a protein-protein interaction. This format enabled us to load the networks with Python and NetworkX to conduct the analysis.

3 Methods

3.1 Structural Characterization

The structural characterization of protein-protein interaction networks provides a first description of how cancer-associated proteins are organized as an interconnected system. In this project, each cancer network is modeled as an undirected graph, where nodes correspond to proteins and edges represent physical or functional interactions. This graph-based representation allows us to move beyond the analysis of individual proteins and study the global architecture of the molecular system associated with each cancer type.

The aim of this section is to define the main topological descriptors used to compare the breast cancer, lung cancer, and ovarian cancer networks. These measures provide information about the size, connectivity, local organization, and distance structure of each network. This initial characterization serves as the basis for later analyses of robustness, functional modularity, spreading dynamics, and the prediction of potential new protein-protein interactions.

3.1.1 Macroscopic Structure

The macroscopic structure of a network enables us to describe its global properties, that is, those properties that summarize how the network is organized as a whole. In the context of protein-protein interaction networks, these measures allow us to understand whether the proteins associated with each cancer type form a compact, highly connected, or modular network, or whether the network is dominated by proteins with a high number of interactions. This information is relevant because the global organization of interactions can influence the stability of the system, the propagation of molecular alterations, and the relative importance of specific proteins within the tumor network.

First, the number of nodes and edges of each network was computed. Nodes represent proteins associated with the analyzed cancer type, while edges represent physical or functional interactions among them. These measures provide a basic description of the size of the network and the number of relationships present between its components. In biological terms, a network with more interactions may reflect a more interconnected molecular system, where alterations in one protein have a higher potential to indirectly affect other related proteins.

The degree of the nodes was also studied, defined as the number of interactions each protein has. From this measure, the minimum, maximum, and average degree of the network were computed. The average degree provides information about the general level of connectivity, while the maximum degree enables the detection of the possible presence of hubs, i.e., highly connected proteins. In PPI networks, these hubs can be especially relevant since they tend to participate in multiple cellular processes or act as integration points between different biological pathways. Therefore, their alteration may have wider effects on the functional organization of the cell.

Another measure analyzed was the clustering coefficient, which quantifies the tendency of the neighbors of a node to be connected to each other. In a PPI network, high clustering may indicate the presence of protein groups that interact in a coordinated manner, possibly associated with protein complexes, signaling pathways, or common biological processes. The minimum, maximum, and average clustering coefficients were computed to evaluate to what extent the network presents a dense local organization. From a biological point of view, these clusters can be relevant since proteins involved in similar molecular functions tend to interact with each other.

Moreover, assortativity was also computed. This metric measures whether nodes tend to connect with other nodes of similar degree. An assortative network indicates that highly connected proteins tend to interact with other highly connected proteins, while a disassortative network suggests that hubs are preferably connected with low-degree proteins. In biological networks, disassortativity is frequent and may have important implications: on one hand, it enables central proteins to connect multiple peripheral proteins; on the other hand, it may cause the network to depend strongly on a small set of hubs.

Finally, metrics related to distances inside the network were also analyzed, such as the average path length and the diameter of the largest connected component. The largest connected component (LCC) corresponds to the largest group of proteins connected to each other through some path inside the network. That is, it represents the main region of the network where all proteins are directly or indirectly connected. This component is used to compute distances because, if the network is not completely connected, not all protein pairs have a path linking them together.

The average path length indicates how many steps, on average, separate two proteins inside the LCC, while the diameter represents the maximum shortest-path distance between two nodes. These metrics allow us to evaluate the efficiency with which a signal, alteration, or functional effect may propagate through the LCC. In the cancer context, a compact LCC with short paths can facilitate communication between proteins and the transmission of molecular perturbations among distinct functional regions.

Apart from the numerical macroscopic descriptors, the degree distribution of each network was also analyzed. This distribution shows how connections are distributed among proteins, that is, how many proteins present a given number of interactions. In PPI networks, this analysis is especially relevant because it enables us to evaluate whether connectivity is distributed homogeneously or, on the contrary, whether there are a few proteins with a very high number of interactions.

To do so, the degree distribution was represented through different graphical approaches. First, the histogram was used to observe the frequency of proteins with each number of interactions. Moreover, the distribution was also represented in logarithmic scale, which facilitates the visualization of heterogeneous patterns and possible long tails, characteristic of many biological networks. These tails indicate that the majority of proteins have few connections, while a few proteins act as highly connected hubs.

Likewise, the complementary cumulative distribution function (CCDF) was studied, which represents the probability of a protein having a degree equal to or higher than a given value. This representation is useful to analyze the tail of the distribution and detect whether the network presents a behavior compatible with a scale-free or power-law network. In these types of networks, hubs have high structural importance, given that they concentrate a significant part of the interactions and can notably influence the stability and functioning of the system.

Finally, an adjustment of the probability density function (PDF) with logarithmic binning was employed to better represent the distribution in regions where there are fewer data points, especially at high degrees.

This procedure reduces noise in the tail of the distribution and enables a clearer comparison of the behavior of the networks. Overall, these plots allow us to visualize the heterogeneity of connectivity and complement the macroscopic metrics, contributing to a more intuitive interpretation of the presence of hubs and the possible hierarchical organization of the PPI networks associated with each cancer type.

3.1.2 Microscopic Structure

The microscopic structure of the network was analyzed with the objective of identifying the relative importance of individual proteins according to their position inside the PPI network. While macroscopic descriptors summarize the global organization of the system, microscopic measures are focused on specific nodes and enable the detection of proteins that may play a crucial structural role. In the context of our networks, this is especially important because some proteins may act as hubs, bridges between functional regions, or central elements in the propagation of molecular alterations.

First, the degree of each node was considered to identify the proteins with the highest number of direct interactions. Proteins with high degree can be interpreted as hubs, given that they interact with many other proteins in the network. These proteins can participate in multiple cellular processes or connect different molecular mechanisms, which makes them potentially relevant elements for the organization of each network.

Moreover, betweenness centrality was also computed to identify proteins that act as bridges inside the network. This metric measures how frequently a node appears in the shortest paths among other node pairs. Therefore, proteins with high betweenness may be important to maintain communication between several regions of the network. In a PPI network, these proteins can be interpreted as possible connection points between functional modules or biological pathways; therefore, their alteration may affect the transmission of information inside the system.

Closeness centrality was used to evaluate to what extent a protein is close to the rest of the network. A protein with high closeness can reach other proteins through relatively short paths, which indicates a central position in terms of accessibility inside the network. These proteins may be relevant in processes where signals or molecular perturbations propagate efficiently through the interaction system.

Moreover, PageRank, eigenvector centrality, and Katz centrality were also computed to capture more global notions of node importance. These measures do not only consider the number of interactions of a protein, but also the importance of the proteins to which it is connected. In this way, a protein may obtain a high score if it is connected to other influential proteins, even if its own degree is not the highest. This is useful in this type of network, where functional relevance may depend not only on direct interactions, but also on the position of the protein inside an important region of the network.

For each centrality metric, the best-ranked proteins were extracted, aiming to compare the most structurally relevant nodes among the different cancer networks. These proteins should not be interpreted directly as biomarkers, but as candidates with possible topological relevance whose biological role may be subsequently discussed or contrasted with biomedical evidence.

3.2 Percolation

To evaluate the structural robustness of PPIs associated with breast, lung, and ovarian cancers, a percolation analysis based on progressive elimination of nodes was performed. This approach allows to study how the global connectivity of a network is degraded in the event of random failures (i.e, eliminations of arbitrary nodes) or targeted attacks (i.e, directed deletions of nodes selected according to some kind of prior knowledge), simulating the loss of proteins or functional alterations within the biological system.

First, the LCC was extracted for each network, with the objective of working only with the main functionally connected part of the graph. Next, nodes were gradually removed following different strategies, evaluating the relative size of the LCC with respect to its original number of nodes after each elimination:

$$S(f) = \frac{|LCC_f|}{N_0}$$

where $S(f)$ represents the relative size of the LCC after eliminating a fraction f of nodes, $|LCC_f|$ is the number of nodes that currently belong to the LCC, and N_0 is the original size of the LCC.

SI DESPRÉS TB FEM ATTACK SEGONS LES PROTEINS QUE LA LITERATURA DIU Q SON IMPORTANTS CALDRÀ AFEGIR-HO AQUÍ. In particular, three different elimination strategies were considered: random failures, degree-based attacks, and betweenness-based attacks. To begin with, in random failures nodes were randomly removed to simulate undirected failures or mutations. Given the stochastic nature of this process, 20 independent replicates were conducted using different seeds, and the mean and standard deviation of the relative size of the LCC were subsequently calculated for each removed fraction. In turn, degree-based attacks consist of eliminating the nodes in decreasing order of degree, starting with the most connected hubs in the network. This strategy models attacks directed at highly interactive proteins and allows to study the network's structural dependence on its central nodes. Finally, betweenness-based attacks eliminate nodes following the decreasing order of betweenness centrality. This analysis allows the identification of critical bridging proteins for maintaining the global connectivity between different functional regions of the network.

For each strategy, the network degradation curve was calculated, representing the relative size of the LCC ($S(f)$) against the fraction of eliminated nodes (f). Additionally, a critical percolation threshold was defined, considering the point at which the giant component fell below 50% of the original size. This value allows for a quantitative comparison of the robustness of the different cancer networks against random disturbances and targeted attacks.

3.3 Community Detection

Community detection is one of the most relevant tools in the analysis of complex networks [4], since it enables the identification of groups of related nodes. These structural results are fundamental for understanding the organization of a system and the underlying patterns that can be found in its interactions. However, community detection is not a trivial problem [5], as different algorithms can produce different partitions depending on the criteria used to define what constitutes a community.

In this work, community detection is approached from multiple theoretical frameworks to ensure robustness and avoid bias towards a single definition of community structure. More preisFive algorithms were applied: Greedy modularity optimization, Louvain [6], Leiden, Infomap and Degree-corrected Stochastic Block Model (DCSBM).

3.4 Epidemic Spreading

4 Results

5 Discussion

6 Conclusion

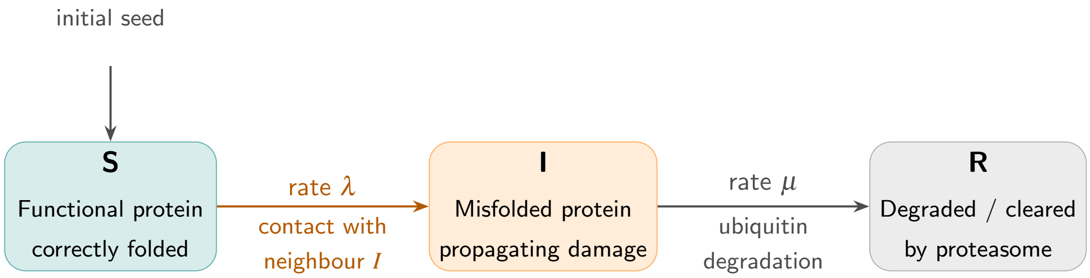

Figure 1: Schematic representation of an SIR-like mathematical model applied to prion-like protein misfolding.

7 References

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